

CS 4720

1) Let us assume that the new Kernel can be represented in the following way:

$$k(x,y) = \alpha k_1(x,y) + (1-\alpha) k_2(x,y), \text{ where } 0 \leq \alpha \leq 1$$

We have been given that k_1 and k_2 are in detailed balance with π , which means that for all states x,y we have $\pi(x)k_1(x,y) = \pi(y)k_1(y,x)$ and $\pi(x)k_2(x,y) = \pi(y)k_2(y,x)$. Now, we need to show that $\pi(x)k(x,y) = \pi(y)k(y,x)$ for all x and y .

Now, let us start by plugging the definition of k into the left hand side, so now we get that $\pi(x)k(x,y) = \pi(x)[\alpha k_1(x,y) + (1-\alpha) k_2(x,y)] = \pi(x)\alpha k_1(x,y) + \pi(x)(1-\alpha) k_2(x,y)$

Now, we can further substitute $\pi(x)k_1(x,y)$ and $\pi(x)k_2(x,y)$ with the values that we figured out earlier in the question:

$$\pi(x)k(x,y) = \alpha\pi(y)k_1(y,x) + (1-\alpha)\pi(y)k_2(y,x) = \pi(y)(\alpha k_1(y,x) + (1-\alpha)k_2(y,x)) = \pi(y)k(y,x)$$

Hence proved that $\pi(x)k(x,y) = \pi(y)k(y,x)$

2) Let us first try to predict the chances of rain before looking at the umbrella on day t through transition model:

$P(R_t = T | \text{umbrellas upto } t-1) = 0.7b_{t-1} + 0.3(1-b_{t-1})$ (which basically comes from the probability of whether it was raining yesterday or not, which is 0.7 if it was raining and 0.3 if it was not raining) $= 0.4b_{t-1} + 0.3$

Now let us use the following formula over here to solve this question:

$$b_t = (P(U_t = T | R_t = T)p_t) / (P(U_t = T | R_t = T)p_t + P(U_t = T | R_t = F)(1-p_t)) \\ = 0.9p_t / (0.9p_t + 0.2(1-p_t))$$

Since, in the long run, the probabilities stop changing, we can say that $b_t = b_{t-1} = x$ and $p_t = y$, so we can say that $y = 0.4x + 0.3$ and now if we update the fixed point equation then it can be represented as:

$$x = 0.9y / (0.9y + 0.2(1-y)) = 0.9y / (0.2 + 0.7y) \\ = 0.9(0.4x + 0.3) / (0.2 + 0.7(0.4x + 0.3)) \\ = 0.9(0.4x + 0.3) / (0.2 + 0.28x + 0.21) \\ = 0.9(0.4x + 0.3) / (0.28x + 0.41) \\ = 0.36x + 0.27 / 0.28x + 0.41 \\ 0.28x^2 + 0.41x = 0.36x + 0.27 \\ 0.28x^2 + 0.05x - 0.27 = 0 \\ 28x^2 + 5x - 27 = 0$$

Let us use the quadratic formula to solve this equation:

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a \text{ (with } a = 28, b = 5 \text{ and } c = -27)$$

$$b^2 - 4ac = 5^2 - 4*(-27)*(28) = 3076$$

So, let us now plugin this value in the formula:

$$x = (-5 + \sqrt{3076}) / 56 \text{ and } (-5 - \sqrt{3076}) / 56 = 50.46/56 \text{ and } -60.46/56 = 0.90 \text{ and } -1.08$$

Now, we got 2 answers over here that 0.90 and -1.08, however, since x is a probability, we know that it can't be negative so $x = 0.90$, hence it means that the probability, given that we observe an umbrella every day is 0.90.

3a) Over here, we have 3 boolean variables at each time which can be represented as follows:

S_t : This tells us whether a student got enough sleep

R_t : This tells us whether a student has red eyes in the class

C_t : This tells us whether a student sleeps in class.

Over here, the question clearly specifies that today's sleep depends on yesterday's sleep, today's red eye depends on today's sleep, and today's sleeping in class depends on today's sleep as well. In this case, our DBN structure has S_t as the hidden state variable and R_t and C_t as the observation variables, and it includes direct edges between S_{t-1} and S_t , S_t and R_t , and S_t and C_t . Also, we have the following conditional probabilities over here:

Prior Probability Table

Variable	Value	Probability
S_1	True	0.7
S_1	False	0.3

Transition Model Table

S_{t-1}	S_t	Probability
T	T	0.8
T	F	0.2
F	T	0.3
F	F	0.7

Sensor Model for Red Eyes

S_t	R_t	Probability
T	T	0.2
T	F	0.8
F	T	0.3
F	F	0.7

Sensor Model for sleeping in class

S_t	C_t	Probability
T	T	0.1
T	F	0.9
F	T	0.3
F	F	0.7

b) Over here, we need to compute the filtered belief, that is, $b_t = P(S_t = \text{True} \mid e_{1:t})$, for $t=1,2$ and 3. Here, each evidence is represented as (R_t, C_t) .

Now, let us compute the filtered belief for $t=1, 2, 3$.

$t = 1$

R_1 and C_1 are false over here, since the student does not have red eyes and is not sleeping in class. Now, we have the following likelihoods:

- $P(e_1 \mid S_1 = T) = 0.8 * 0.9 = 0.72$
- $P(e_1 \mid S_1 = F) = 0.3 * 0.7 = 0.21$

Now, we have the following unnormalized posteriors:

$$0.72 * 0.7 = 0.504$$

$$0.21 * 0.3 = 0.063$$

Now, we can calculate the normalized posteriors:

$$b_1 = 0.504 / (0.504 + 0.063) = 0.889$$

t=2

R_2 is true and C_2 is false over here, since the student has red eyes and is not sleeping in class. Now, we have the following likelihoods:

- $P(e_2 | S_2 = T) = 0.2 * 0.9 = 0.18$
- $P(e_2 | S_2 = F) = 0.7 * 0.7 = 0.49$

Now, we have the following unnormalized posteriors:

$$0.18 * 0.745 = 0.1341$$

$$0.49 * 0.255 = 0.12495$$

Now, we can calculate the normalized posteriors:

$$b_2 = 0.1341 / (0.1341 + 0.12495) = 0.518$$

t=3

R_3 and C_3 are true over here, since the student does have red eyes and is sleeping in class. Now, we have the following likelihoods:

- $P(e_3 | S_3 = T) = 0.2 * 0.1 = 0.02$
- $P(e_3 | S_3 = F) = 0.7 * 0.3 = 0.21$

Now, we have the following unnormalized posteriors:

$$0.02 * 0.559 = 0.01118$$

$$0.21 * 0.441 = 0.09261$$

Now, we can calculate the normalized posteriors:

$$b_3 = 0.01118 / (0.01118 + 0.09261) = 0.108$$